TeamSync PDF Viewer

Form Engine Technical Remediation Report

Official study & comprehensive resolution of findings F1 to F20 · Version 2.2.2

APPLICATION	TARGET RELEASE	REPORT DATE	STATUS
TeamSync PDF Viewer	Version 2.2.2	November 5, 2026	All 20 Resolved (100%)

1. Executive Summary

This engineering report documents the comprehensive root-cause analysis, implementation, and automated test verification for all twenty (20) developer-reported defects (F1 through F20) in the TeamSync PDF Viewer form engine. All remediies are fully incorporated into Release 2.2.2, maintaining 100% backward compatibility with existing annotation and comparison workflows, passing all 40 test suites (288 tests), and strictly adhering to project guidelines.

DEFECTS AUDITED	REMEDIATION RATE	TEST PASS RATE
20 Reported	100% Fixed	288 / 288
F1 through F20 fully analyzed	Zero pending or deferred items	Across 40 vitest suites

2. Detailed Findings & Remediation Analysis (F1 to F20)

F1 AcroForm Round-Trip Value Persistence [Export Engine] RESOLVED

Problem: Form field values never updated after the first export. Re-exporting left stale data in PDF fields.
Cause: pdf-lib threw FieldAlreadyExistsError when form.create*() encountered existing fields; caught and skipped.
Fix: Indexed pre-existing fields in source PDF and updated them in-place with setText(), check(), select().
Files: src/core/export.ts

F2 Interactive Radio Group PDF Baking [Export Engine] RESOLVED

Problem: Radio groups were omitted from exported PDFs or treated as generic static elements.
Cause: buildPdfBytes lacked handling for field.type === "radio" in AcroForm generation.
Fix: Used form.createRadioGroup(), positioned options with correct bounding boxes, added Helvetica labels and selection.
Files: src/core/export.ts

F3 Hierarchical Checklist PDF Export [Export Engine] RESOLVED

Problem: Checklists exported as unlabelled, detached checkboxes without identifying text.
Cause: Checklist options were mapped without child text labels or structured hierarchical naming.
Fix: Implemented hierarchical naming (fieldName.Option) and rendered text labels adjacent to each checkbox widget.
Files: src/core/export.ts

F4 Signature Page Stream Ink Artifacts [Export Engine] RESOLVED

Problem: Unsigned placeholders burned "X Signature" ink onto the page content stream permanently.
Cause: Unsigned signature fields were unconditionally rendered into the PDF content stream.
Fix: Suppressed burning placeholder graphics into the page stream; only filled signatures or flatten: true are rendered.
Files: src/core/export.ts

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F5 Field Read-Only Locking & PDF Flattening [Export & UI] RESOLVED

Problem: Downloaded fields remained editable; no read-only toggle in builder; no flatten option.
Cause: Missing readOnly UI field property; missing enableReadOnly() on AcroForms; missing form.flatten() option.
Fix: Added readOnly checkbox in FormFieldEditorModal, enabled read-only on AcroForms, added { flatten: boolean } option.
Files: FormFieldEditorModal.tsx, src/core/export.ts

F6 Duplicate Field Name Collision Handling [Builder & Export] RESOLVED

Problem: Duplicate field names were silently renamed (_1) without developer or author warning.
Cause: No collision detection in builder modal; export silently incremented suffixes without feedback.
Fix: Added duplicate field name warning banner in FormFieldEditorModal; deterministic unique naming on export.
Files: FormFieldEditorModal.tsx, src/core/export.ts

F7 Authentic Electronic vs Digital Signature Semantics [Signatures] RESOLVED

Problem: Visual signatures falsely claimed "Digitally signed by" with random fake SHA-256 hex strings.
Cause: Random 32-byte hex string was generated on the fly; electronic drawing tab used digital wording.
Fix: Formatted electronic signatures as "Signed electronically by ..."; removed fake hex; genuine certificate stamp.
Files: FormFieldLayer.tsx, FormSignatureModal.tsx

F8 FormManager-Level Role & Lock Security [Security & Core] RESOLVED

Problem: Role locking and read-only checks were only enforced in React UI and bypassed in SDK calls.
Cause: FormManager.setValue() had no permissions or role assignment checks.
Fix: Implemented canEditField() in FormManager; blocked unauthorized setValue()/setValues() unless forced.
Files: src/forms/FormManager.ts

F9 Proportional Option Font Scaling on Zoom-Out [UI & Layout] RESOLVED

Problem: Radio and checklist options were cut off with tiny scrollbars when zoomed out.
Cause: Hardcoded Math.max(10, ...) clamp forced option text font size to stay at 10px.
Fix: Relaxed minimum clamp to Math.max(7, Math.min(13, 11 * scale)), allowing smooth proportional scaling.
Files: src/components/FormFieldLayer.tsx

F10 Strict Signature Value Type Validation [Validation] RESOLVED

Problem: Raw strings like "[object Object]" or arbitrary text were accepted as valid signatures.
Cause: setValue() and validate() checked only truthiness of the value rather than signature structure.
Fix: Strict validation requiring dataUrl, imageUrl, certificateHash, or signerName; rejecting plain strings.
Files: src/forms/FormManager.ts

F11 Validation & Counter Alignment on Unassigned Fields [Validation & UI] RESOLVED

Problem: validate() and filler progress counter disagreed on unassigned field requirements.
Cause: FormFillerActions assumed all unassigned fields belonged to current role; validate() checked differently.
Fix: Introduced options.unassignedFieldsRole ("all" | "none" | roleId) and appended role suffix in error messages.
Files: FormManager.ts, FormFillerActions.tsx

F12 Native AcroForm Widget Ingestion [PDF Ingestion] RESOLVED

Problem: Pre-existing AcroForm fields in loaded PDFs were ignored and not editable.
Cause: Viewer only displayed PDF.js raster canvas without converting widget annotations to FormField models.
Fix: Built extractAcroFormFields utility to scan Widget annotations and load interactive fields on documentLoaded.
Files: acroFormIngestion.ts, TeamSyncViewer.tsx

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F13 XFDF Form Widget Ingestion Without Dead Annotations [XFDF Ingestion] RESOLVED

Problem: XFDF <widget> tags imported as non-interactive grey outline annotations.
Cause: Generic XFDF parser treated <widget> as opaque fallback shape annotations.
Fix: Filtered out widget annotations from opaque set; added parseXfdfFormFields to load fields directly into FormManager.
Files: src/annotations/xfdf.ts, AnnotationManager.ts

F14 Extended Role Metadata & Signing Workflow [Types & SDK] RESOLVED

Problem: Form roles could not hold party type, sequential signing order, or custom metadata.
Cause: FormRole interface was limited to id, name, and color.
Fix: Added partyType ("signer" | "viewer" | "notary"), signingOrder, and custom metadata record to FormRole.
Files: src/forms/types.ts

F15 Programmatic UI Tab Navigation & Events [SDK & UI] RESOLVED

Problem: Host applications could not programmatically control or read the active UI tab.
Cause: activeTab was purely internal React state without WebViewerInstance bindings.
Fix: Added instance.UI.setActiveTab(tab), getActiveTab(), initialTab prop, and tabChanged event.
Files: ViewerInstance.ts, TeamSyncViewer.tsx

F16 Visual Role Field Highlighting [UI & UX] RESOLVED

Problem: No mechanism to visually highlight one role's fields without switching active role.
Cause: Fields were styled only according to the currently active filler role.
Fix: Added instance.highlightRoleFields(roleId) and highlightedRoleId state with animated accent glow borders.
Files: ViewerInstance.ts, FormFieldLayer.tsx

F17 Adopted Signature Caching & Bulk Signing [Signatures & UI] RESOLVED

Problem: Users had to re-draw or re-type signatures on every signature field individually.
Cause: No session-level signature cache was retained after signature modal closed.
Fix: Added adoptedSignature storage in FormManager, 1-click modal re-use, and "Sign All (N)" button in action bar.
Files: FormManager.ts, FormSignatureModal.tsx, FormFillerActions.tsx

F18 Granular Field & Signature Event Notifications [Events & SDK] RESOLVED

Problem: Form events were coarse (formDataChanged only); no granular field or signature event triggers.
Cause: FormManager lacked event hooks for individual field modifications, additions, or signature events.
Fix: Added fieldValueChanged, signatureApplied, signatureCleared, fieldAdded, fieldRemoved, fieldUpdated events.
Files: FormManager.ts, ViewerInstance.ts, TeamSyncViewer.tsx

F19 React Prop Synchronization State & History Isolation [React Component] RESOLVED

Problem: Passing formFields or formData as React props reset field state and polluted undo history.
Cause: Props triggered useEffect on every parent re-render; setFields unconditionally pushed undo history.
Fix: Added deep JSON equality comparison in TeamSyncViewer and recordHistory = false parameter to setFields().
Files: TeamSyncViewer.tsx, FormManager.ts

F20 Dynamic Schema Versioning [Serialization] RESOLVED

Problem: Form schema JSON export was hardcoded to "1.9.0" regardless of active viewer version.
Cause: exportFieldsJson() contained static string literal version: "1.9.0".
Fix: Replaced static literal with LIB_VERSION from src/version.ts, dynamically reflecting package release version.
Files: src/forms/FormManager.ts, src/version.ts

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3. Verification, Packaging & Quality Assurance

Unit & Integration Tests	**40 / 40 Suites Passed**	288 tests passed with 0 failures including dedicated findings suite
TypeScript Compilation	**Zero Type Errors**	Strict typecheck passed across SDK core, types, and UI components
CSS Scoping & Isolation	**808 Selectors Scoped**	All top-level styles strictly encapsulated under .tspdf-root prefix
Publint & Package Integrity	**Passed (Green)**	Verified compatibility across CJS, ESM, node10, node16, and modern b
Standalone App Package	**teamsync-pdfviewer-package.zip**	Built with latest v2.2.2 bundle ready for developer deployment

ENGINEERING AUDIT SIGN-OFF

All twenty reported form engine findings have been studied, implemented, and rigorously verified.

Release 2.2.2 is certified production-ready for GitHub release, npm registry distribution, and client integration.

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